12th BOARD HACKER JUGAAD NOTES :-

CALCULUS

Calculus is a branch of mathematics that involves the study of rates of change

CONCEPT OF CALCULUS

The main concepts of Calculus are :

- Limits
- Differential Calculus (Differentiation)
- Integral Calculus (Integration)
- Multivariable Calculus (Function theory).

INDETERMINANT FORMS

An undefined expression involving some operation between two qualities is called an indeterminate form if it does not evaluate to a single number value or infinity.

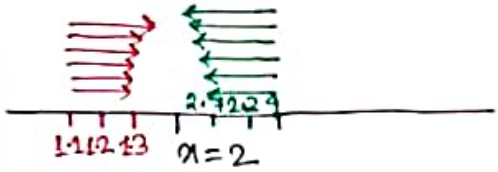
$$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty, \infty^0, 0^0, 1^\infty$$

LIMITS

A Limit is defined as a value that a function approaches the output for the given input values.

$\lim_{x \to a} f(x) = C$ or $\lim_{x \to 2} (x+3)$

If x is approaching towards a than $f(x)$ is approaching towards C.



$f(x) = x + 3$

Limit $\to 5$

$x = 1.1 \quad f(1.1) = 1.1 + 3 = 4.1$

$x = 1.2 \quad f(1.2) = 1.2 + 3 = 4.2$

$x = 1.3 \quad f(1.3) = 1.3 + 3 = 4.3$

$x = 1.4 \quad f(1.4) = 1.4 + 3 = 4.4$

$x = 1.5 \quad f(1.5) = 1.5 + 3 = 4.5$

$x = 1.9 \quad f(1.9) = 1.9 + 3 = 4.9$

$x = 2 \quad f(2) = 2 + 3 = 5$

$f(x) = x+3$

$x = 2.9 \quad f(2.9) = 2.9+3 = 5.9$

$x = 2.8 \quad f(2.8) = 2.8+3 = 5.8$

$x = 2.7 \quad f(2.7) = 2.7+3 = 5.7$

$x = 2.6 \quad f(2.6) = 2.6+3 = 5.6$

$x = 2.5 \quad f(2.5) = 2.5+3 = 5.5$

$x = 2.1 \quad f(2.1) = 2.1+3 = 5.1$

$x = 2 \quad f(2) = 2+3 = 5$

R.H.L

$\lim_{x\to a} f(x) = C$

L.H.L $\lim_{x\to a^-} f(x)$

$x = a-h$, $h \to 0$

$\lim_{h\to 0} f(a-h) = C$

R.H.L $\lim_{x\to a^+} f(x)$

$x = a+h$, $h \to 0$

$\lim_{h\to 0} f(a+h) = C$

(1) $\lim_{x\to 0} f(x) = C$

If L.H.L = R.H.L

$\Rightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x)$

$\Rightarrow$ then only limit exist

(2) If L.H.L $\neq$ R.H.L, then $\lim_{x\to a} f(x)$ does not exist.

$\lim_{x\to 2} (x+3)$

L.H.L $\lim_{x\to 2^-} (x+3) = \lim_{h\to 0} (2-h+3) = \lim_{h\to 0} (5-h)$

$= 5-0 = 5$

$x = 2-h$, $h \to 0$

R.H.L $\lim_{x\to 2^+} (x+3) = \lim_{h\to 0} (2+h+3) = \lim_{h\to 0} (5+h) = 5$

$x = 2+h$, $h \to 0$

L.H.L = R.H.L

$\Rightarrow$ limit exist

$\Rightarrow \lim_{x\to 2} (x+3) = 5$ Ans.

Properties of Limits

1. $\lim_{x\to a} [f(x)+g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$

2. $\lim_{x\to a} [f(x)-g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$

3. $\lim_{x\to a} [f(x)\cdot g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$

4. $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$

5. $\lim_{x\to a} [kf(x)] = k\left[\lim_{x\to a} f(x)\right]$

$\lim_{x\to 2} [3x^2 + 5x^3]$

$\Rightarrow \lim_{x\to 2} 3x^2 + \lim_{x\to 2} 5x^3$

$\Rightarrow 3 \times \lim_{x\to 2} (x^2) + 5 \lim_{x\to 2} (x)^3$

CONTINUITY

A function $f(x)$ is said to be continuous at $x=a$

$$\lim_{x\to a} f(x) = f(a)$$

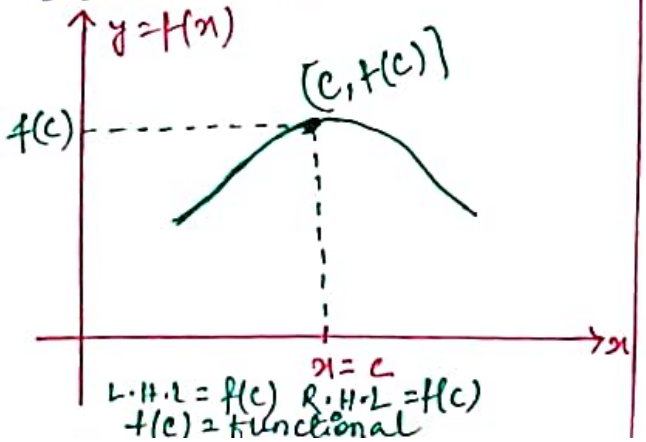
(Limiting value) (functional value)

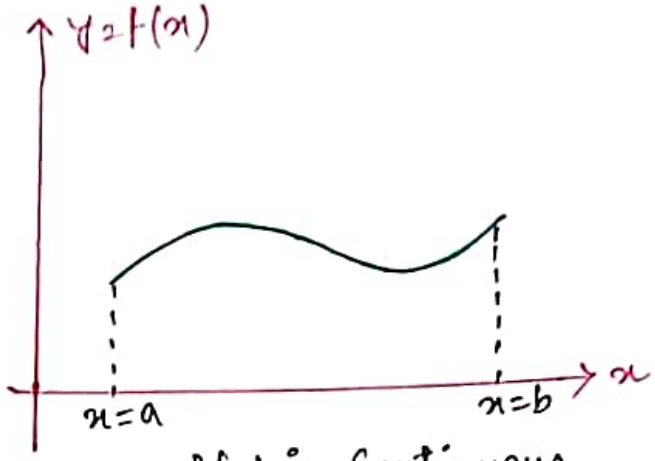
$$\lim_{x\to a} f(x) = \lim_{x\to a} f(x) = f(a)$$

GEOMETRICAL MEANING OF CONTINUITY

i) Function f will be continuous at $x=c$ if there is no break in the graph of the function at the point $[c, f(c)]$.

ii) In an interval, function is said to be continuous if there is no break in the graph of the in the entire interval.





f(x) is Continuous

Algebra of Continuous functions Continuity

① All the polynomial functions are always continuous on all Real values

e.g. $f(x) = 3x^5 - 4x^3 + 7x^2 + 2x + 8$

② All the algaebraic functions are always Continuous in their domain.

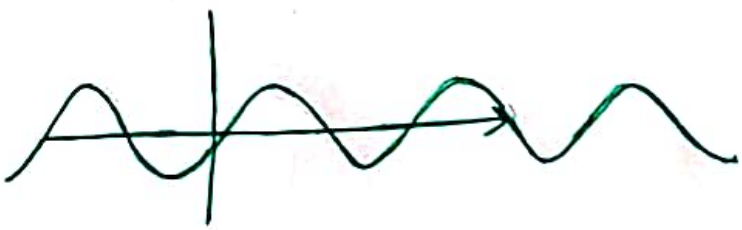
$$f(x) = \frac{x^2-5}{x+2} \qquad x+2 \neq 0$$

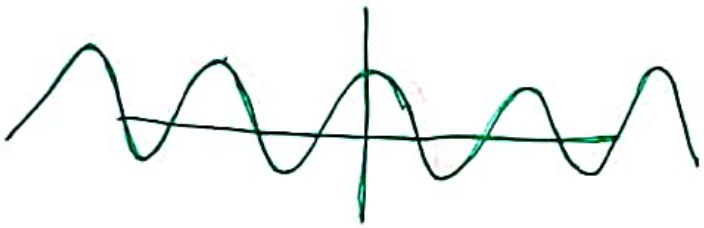
$$x \neq -2$$

$$x \in R - \{2\}$$

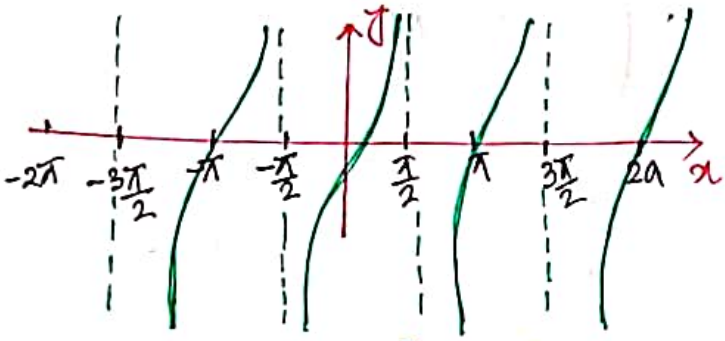
$f(x)$ is always Continuous $R-\{-2\}$

③ Sin x & Cos x functions are always continuous on Real values.





④ $\tan x$ is not continuous at $x = (2n+1)\frac{\pi}{2}$



Continuous Functions

If $f(x)$ & $g(x)$ are two continuous functions at $x=c$, then

① $f(x) + g(x)$ is also continuous at $x=c$

② $f(x) - g(x)$ is also continuous at $x=c$

③ $f(x) \times g(x)$ is also continuous at $x=c$

④ $f(x) / g(x)$ is also continuous at $x=c$ but $g(c) \neq 0$

Continuity of Composite Functions

If $f(x)$ & $g(x)$ are continuous functions at $x=c$, then their composition is always continuous at $x=c$

$fog(x)$ & $gof(x)$ are always continuous at $x=c$

Ex. $\sin(x^5)$

$\sin x$ ← x^5

$\Rightarrow$ $\sin x$ is always continuous on R

$\Rightarrow$ x^5 is a polynomial which is always continuous R then By theorm, $\sin(x^5)$ is also continuous.

DISCONTINUITY

A function $f(x)$ is said to be discontinuous at $x=a$ if

1. $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ exist, but are not equal.
2. $\lim_{x \to a^-} f(x)$ and $\lim_{x \to a^+} f(x)$ exist and are equal but not equal to $f(a)$.
3. $f(a)$ is not defined.
4. At least one of the limits does not exist.

DIFFERENTIABILITY

If a function $f(x)$ is differentiable at $x = a$, then

$$\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \text{ exist}$$

L.H.L $\quad$ R.H.L

$$\lim_{x \to a^-} \frac{f(x) - f(a)}{x - a} \qquad \lim_{x \to a^+} \frac{f(x) - f(a)}{x - a}$$

L.H.D (Left Hand Derivative) $\quad$ Right Hand derivatives (RHD)

If L.H.D = R.H.D, then only function $f(x)$ is differentiable.

If LHD $\neq$ RHD, then $f(x)$ is not differentiable.

L.H.D Left Hand Derivative

$$\lim_{x \to a^-} \frac{f(x) - f(a)}{x - a}$$

$x = a - h$, $h \to 0$

$$\lim_{h \to 0} \frac{f(a-h) - f(a)}{a - h - a}$$

$$\Rightarrow \lim_{h \to 0} \frac{f(a-h) - f(a)}{-h}$$

R.H.D (Right Hand Derivative)

$$\lim_{x \to a^+} \frac{f(x) - f(a)}{x - a}$$

$x = a + h$, $h \to 0$

$$= \lim_{h \to 0} \frac{f(a+h) - f(a)}{a + h - a}$$

$$= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

LHD = RHD = $f(a)$ = Derivative of $f(x)$ $a + x = a$

Differentiation of $f(x)$ $a + x = a$

h-Method

If $f(x)$ is differentiable at $x = a$, then $\lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ exist

$x - a = h$

$x = a + h$

$h \to 0$

$$\lim_{h \to 0} \frac{f(a+h) \cdot f(a)}{a + h - a}$$

$$\boxed{f(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}}$$

Derivative of $f(x)$ at $x = a$

DERIVATIVES OF STANDARD FUNCTIONS

1. $\frac{d}{dx}(\sin x) = \cos x$
2. $\frac{d}{dx}(\tan x) = \sec^2 x$
3. $\frac{d}{dx}(\cos x) = -\sin x$
4. $\frac{d}{dx}(\sec x) = \sec x \tan x$
5. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
6. $\frac{d}{dx}(\operatorname{cosec} x) = -\cot x \operatorname{cosec} x$
7. $\frac{d}{dx}(\text{constant}) = 0$
8. $\frac{d}{dx}(x^n) = n \times x^{n-1}$

9. $\frac{d}{dx}(e^x) = e^x$

10. $\frac{d}{dx}(\ln x) = \frac{1}{x}, x > 0$

DERIVATIVES OF STANDARD FUNCTIONS

11. $\frac{d}{dx}(a^x) = a^x \ln a$

12. $\frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$

13. $\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$

14. $\frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$

15. $\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$

16. $\frac{d}{dx}(\sec^{-1}x) = \frac{1}{|x|\sqrt{x^2+1}}$

17. $\frac{d}{dx}(\operatorname{cosec}^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$

$\frac{dy}{dx}$ = differentiate y with respect to x

$\frac{d(\sin x)}{dx} = \cos x$ ✓

$\frac{d(\sin x)}{dy} = \cos x$ ✗

Basic rules

1. $\frac{d}{dx}[kf(x)] = k\frac{d}{dx}[f(x)]$, $k \in$ Constant

$\frac{d(3\sin x)}{dx} = 3\frac{d(\sin x)}{dx} = 3\cos x$

2. $\frac{d}{dx}[u(x) + v(x)] = \frac{d}{dx}u(x) + \frac{d}{dx}v(x)$

$\frac{d(\sin x + \tan x)}{dx} = \frac{d(\sin x)}{dx} + \frac{d(\tan x)}{dx}$

$= \cos x + \sec^2 x$

3. Product rule

$\frac{d}{dx}u(x)\cdot v(x) = u(x)\frac{d}{dx}v(x) + v(x)\frac{d}{dx}u(x)$

4. QUOTIENT RULE

$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$

PRODUCT RULE :-

$y = f(x)\cdot g(x)$

$\frac{dy}{dx} = \frac{d[f(x)\cdot g(x)]}{dx} = f(x)\cdot\frac{d[g(x)]}{dx} + g(x)\times\frac{d[f(x)]}{dx}$

$\Rightarrow \frac{d(\sin x \times \tan x)}{dx} = \sin x \times \frac{d(\tan x)}{dx} + \tan x \times \frac{d(\sin x)}{dx} = \sin x \times \sec^2 x + \tan x \times \cos x$

QUOTIENT RULE :-

$y = \frac{N}{D} = \frac{f(x)}{g(x)}$

$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{N}{D}\right) = \frac{D\times\frac{dN}{dx} - N\times\frac{dD}{dx}}{D^2}$

$y = \frac{\sin x}{\tan x}$

$\frac{dy}{dx} = \frac{\tan x \times \frac{d(\sin x)}{dx} - \sin x \times \frac{d(\tan x)}{dx}}{(\tan x)^2}$

$\frac{dy}{dx} = \frac{\tan x \cos x - \sin x \sec^2 x}{(\tan x)^2}$

$x^4 = a \times x^{n-1}$

① $\frac{d(x^5)}{dx} = 5 \times x^{5-1} = 5 \times x^4 = 5x^4$

② $\frac{d(5)}{dx} = 0$

③ $\frac{d(5\sin x)}{dx} = \frac{5 \times d(\sin x)}{dx} = 5\cos x$

④ $\frac{d(x)}{dx} = 1 \times x^{1-1} = 1 \times x^0 = 1$

⑤ $\frac{d(\sqrt{x})}{dx} = \frac{d(x^{1/2})}{dx} = \frac{1}{2} \times x^{\frac{1}{2}-1}$

$= \frac{1}{2} \times x^{-1/2}$

$= \frac{1}{2} \times \frac{1}{x^{1/2}}$

$= \frac{1}{2} \times \frac{1}{\sqrt{x}} = \frac{1}{2\sqrt{x}}$

DERIVATIVES OF COMPOSITE FUNCTIONS

CHAIN RULE

Power ↓ outer ↓ inner

$f(x) = \sin(x^3)$

$f'(x) = \cos(x^3) \times 3x^2$

$= 3x^2 \cos(x^3)$ Ans.

$f(x) = \sin^2(x^3)$

$= [\sin(x^5)]^2$

$= 2[\sin(x^5)] \times \cos(x^5) \times 5x^4$ Ans

DERIVATIVES OF IMPLICIT FUNCTIONS

$\frac{d(\sin y)}{dx} = \cos(y) \times \frac{dy}{dx}$

$\frac{d(\sin y^3)}{dx} = \cos(y^3) \times 3y^2 \times \frac{dy}{dx}$

$\frac{d(\sin x)}{dx} = \cos x$

$\frac{d(\sin y)}{dx} = \cos y \times$

$\cos x \times$

LOGARITHM

$f(x) = \log_a x$ [log of x to the base a]

Argument $= x > 0$

Base $= a > 0,\ a \neq 1$

1. $\log a^a = 1$ ex $\log 8^8 = 1$
2. $\log a^1 = 0$ ex, $\log 8^1 = 0$
3. $\log_b a = \frac{\log a}{\log b}$ ex, $\log_3 7 = \frac{\log 7}{\log 5}$
4. $\log_b a = \frac{1}{\log_a b}$ ex, $\log_5 7 = \frac{1}{\log_7 5}$
5. $\log_b a + \log_b c = \log_b (a \times c)$

ex, $\log_3 5 + \log_3 7 = \log_3 (5 \times 7)$

$= \log_3 35$

6. $\log_b a - \log_b c = \log_b \left(\frac{a}{c}\right)$

ex, $\log_3 5 - \log_3 7 = \log_3 \left(\frac{5}{7}\right)$

7. $\log_b a^c = c \times \log_b a$

ex, $\log_4 5^3 = 3 \times \log_4 5$

LOGARITHMIC DIFFENRENTIATION

1: $y = f_1(x) \cdot f_2(x) \cdot f_3(x) \cdot f_4(x)$

$\log y = \log [f_1(x) \cdot f_2(x) \cdot f_3(x) \cdot f_4(x)]$

$\log y = \log f_1(x) + \log f_2(x) + \log f_3(x) + \log f_4(x)$

2. $y = [f(x)]^{g(x)}$

$\log y = \log [f(x)]^{g(x)}$

$\log y = g(x) \times \log [f(x)]$

DERIVATIVES OF FUNCTIONS IN PARAMETRIC FORMS

$x = 2\cos\theta \quad y = \sin\theta \quad \frac{dy}{dx} = ?$

$\frac{dx}{d\theta} = 2(-\sin\theta) \quad \frac{dy}{d\theta} = \cos\theta$

$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\cos\theta}{-2\sin\theta}$ Ans.

SECOND ORDER DERIVATIVE

$y = f(x)$

$\frac{dy}{dx} = f'(x)$

$\frac{d}{dx}\left(\frac{dy}{dx}\right) = f''(x)$

$$\boxed{\frac{d^2y}{dx^2} = f''(x)}$$

$\frac{d^2y}{dx^2} = f''(x)$

$= y'' = y_2$

2) Second order derivati -ves Double differentiation

ex $y = 3\sin x$

$\frac{dy}{dx} = 3\cos x$

$\frac{d^2y}{dx^2} = 3(-\sin x)$

$= -3\sin x$ Ans.